# Software Configuration Management Document

PetroPulse - Real-Time Sentimental Analytics Dashboard

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Submitted for the course
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# Contents

**1 Software Configuration Management (SCM) 3**

1.1 Introduction . . . . . 3

1.2 Objectives . . . . . 3

1.3 Configuration Items (CIs) . . . . . 3

1.4 Version Control . . . . . 3

1.5 Change Management Process . . . . . 4

1.6 Configuration Status Accounting . . . . . 4

1.7 Configuration Audits . . . . . 4

1.8 Roles and Responsibilities . . . . . 5

**2 SCM Process Flow 5**

**3 Conclusion 6**

# 1 Software Configuration Management (SCM)

## 1.1 Introduction

Software Configuration Management (SCM) is a critical process that ensures the integrity and consistency of a software system by managing changes in a controlled and traceable manner. For PetroPulse, which integrates real-time data ingestion, NLP-based sentiment analysis, and dashboard visualization, SCM plays a vital role in maintaining system stability and reliability.

Due to the dynamic nature of APIs, machine learning models, and frequent updates in data processing pipelines, it is essential to maintain strict version control and documentation practices.

## 1.2 Objectives

- Maintain system consistency and integrity
- Track changes in source code and documents
- Enable collaboration among team members
- Ensure proper documentation of changes
- Ensure reproducibility of builds

## 1.3 Configuration Items (CIs)

- Software Requirements Specification (SRS)
- Software Design Document (SDD)
- Source Code (Frontend + Backend + NLP Models)
- Database Schema (PostgreSQL/MongoDB)
- Test Cases and Reports
- API Configurations (Twitter/X, NewsAPI)

## 1.4 Version Control

Version control is maintained using Git and GitHub.

**Tool Used:** Git and GitHub

**Repository:** PetroPulse Repository

**Branches:**

- Main branch : Stable Production code.

- Development branch
- Feature branches (Data ingestion, sentiment analysis, dashboard.)

**Version Naming:**

- v1.0 – Basic data ingestion
- v2.0 – Sentiment analysis integration
- v3.0 – Dashboard and visualisation

## 1.5 Change Management Process

The system follows a structured change management process:

1. A change request is submitted
2. Impact analysis is performed
3. Approval is given by the project manager
4. Changes are implemented in a feature branch
5. Testing is performed
6. Changes are merged and version updated

## 1.6 Configuration Status Accounting

Configuration status accounting involves recording and tracking all changes made to configuration items. Git logs and commit messages provide: Detailed change history, Bug fix tracking, Feature enhancement tracking, Version mapping. This ensures transparency and traceability across development phases.

## 1.7 Configuration Audits

Configuration audits ensure that the system complies with defined requirements.

**Functional Audit**

- Verifies that all functionalities such as real-time ingestion, sentiment classification, and dashboard visualization are implemented correctly.

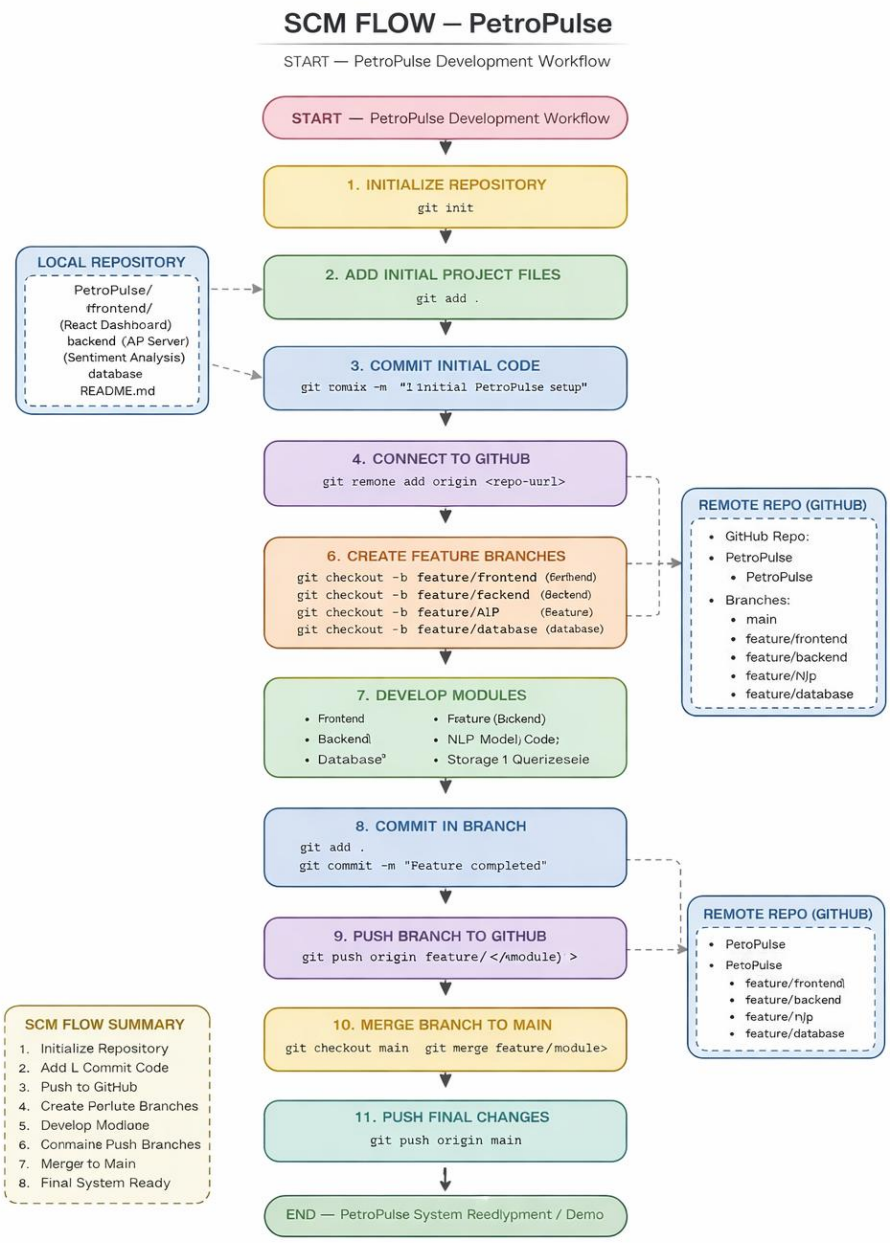
**Physical Audit**

- Ensures all artifacts (code, documents, test cases) are properly stored and versioned.

### 1.8 Roles and Responsibilities

- **SCM Manager** – Oversees version control and configuration process.
- **Developers** – Implements features and push updates.
- **Testers** – Validates system correctness.
- **Project Lead** – Approves changes and monitors progress

# 2 SCM Process Flow




# 3 Conclusion

The SCM framework ensures efficient change handling, improved collaboration, and enhanced software quality. It enables PetroPulse to evolve continuously while maintaining system integrity and reliability.